ENGINEERING PORTFOLIO · OCTOBER 2026

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B.S. Electrical & Computer Engineering, intended additional major in Robotics · May 2028

Robotics software test engineering intern at **Locus Robotics**, building hand-eye calibration checks and Gazebo simulations for Array, the picking robot in this photo. At CMU I write C++ motion planners for 7-DoF arms and a bare-metal Cortex-M4 kernel. My MIT CSAIL work on 3D-printed motion sensing was published at CHI '26.

Applying for Summer 2027 internships in robotics and embedded software.

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Locus Array. Photo: Locus Robotics.

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Contents, mapped onto the robot stack from logic gates up to test tooling

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3	**Hand-eye calibration checks for Locus Array**	Internship, Locus Robotics		●	●		●
5	**Decentralized planner for in-space assembly**	16-891, graduate course				●	●
6	**Two-arm pick, handoff, and place**	16-350 course project				●	●
7	**Bare-metal Cortex-M4 kernel**	18-349 course project		●	●		
8	**FPGA game logic in SystemVerilog**	18-240 course project	●				
9	**Xspine application interfaces**	MIT CSAIL, published CHI '26		●	●		
10	**Teaching and teams**	CMU					

LANGUAGES	C++, Python, C, SystemVerilog, ARM assembly, MATLAB, JavaScript, Bash
ROBOTICS	ROS 2, Gazebo, OpenCV, PyBullet, Docker, Linux, Git, CI/CD, GDB, pytest, Grafana/VictoriaMetrics, A*, RRT/PRM, MAPF
HARDWARE	STM32 (Cortex-M4), FPGA (Vivado), NVIDIA Orin, UART/I2C/ADC, PCB design, Arduino, Raspberry Pi, SOLIDWORKS, Fusion 360, oscilloscope, soldering

A hand-eye calibration check that finds drift without recalibrating the robot

Array picks items with a 6-DoF arm guided by a multi-camera rig. If the cameras and the arm's own kinematics disagree about where the gripper is, the arm reaches for the wrong spot.

I built the ROS 2 tooling that measures that disagreement and shows where it comes from.

WHAT I BUILT

- Hand-eye calibration validation: fiducials and PnP give the cameras' estimate of the arm pose, forward kinematics gives the arm's own, and TF composition lines the two up
- Reprojection-error and point-cloud overlay visualizers in OpenCV
- Runs on Hardware-in-the-Loop benches with NVIDIA Orins

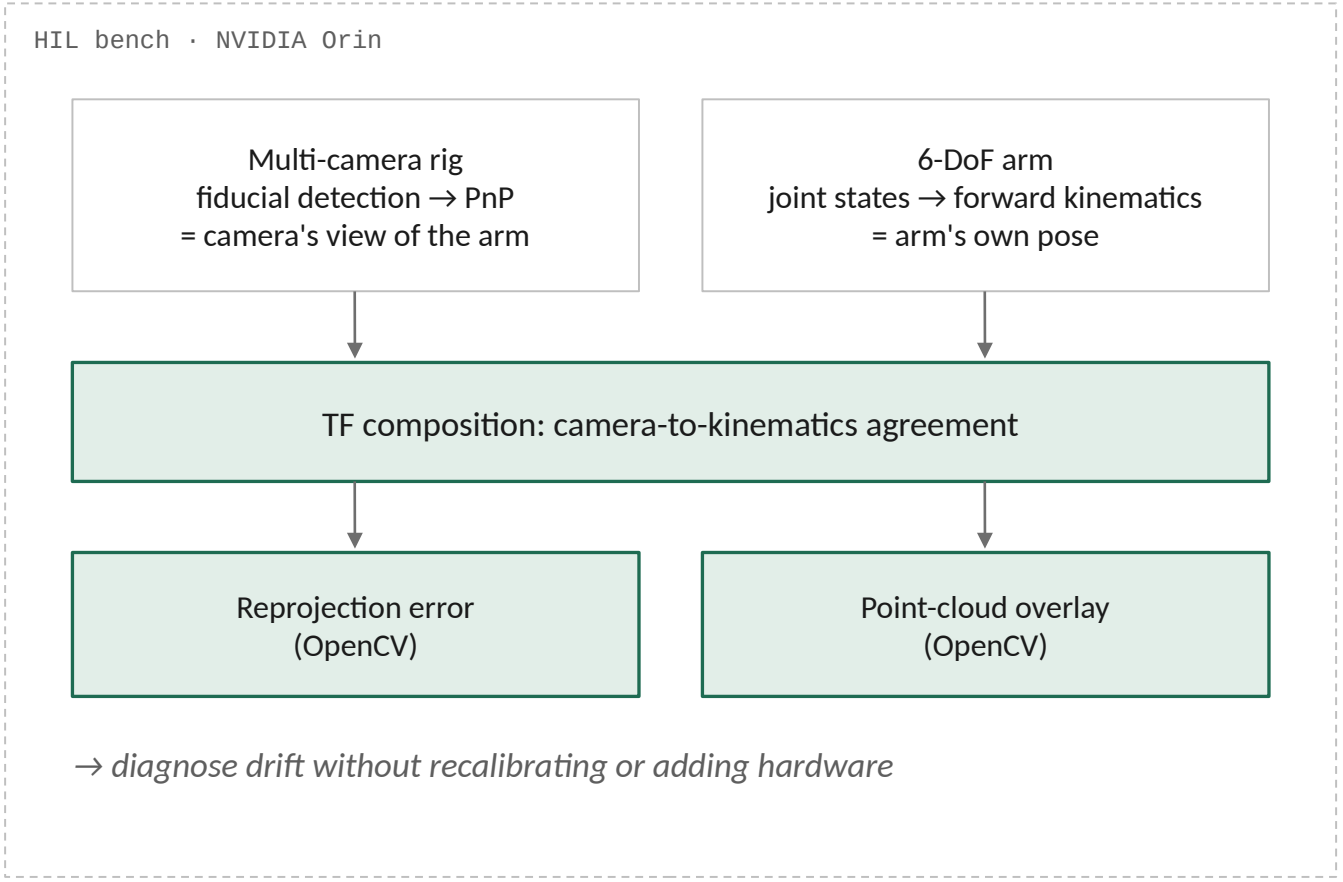
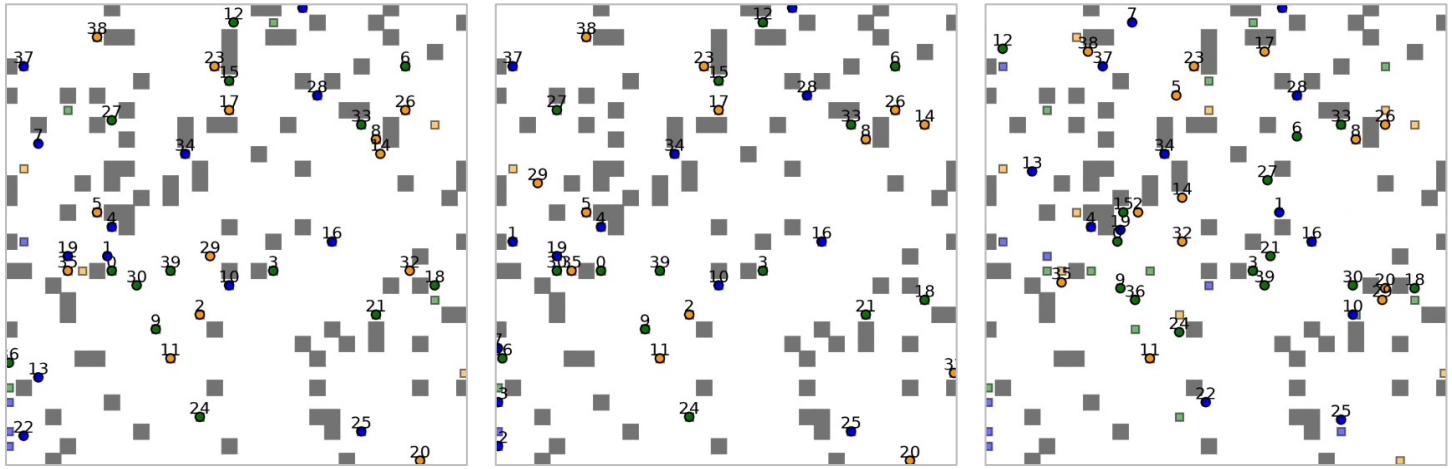


fig. 1 Calibration validation pipeline on the HIL bench. built by me

Besides calibration, I put the picking system into Gazebo and wired fleet data into Grafana

Gazebo	Integrated the arm and camera rig into Gazebo simulation, so picking-system work can happen without a robot.
Grafana / VictoriaMetrics	Dashboards tracking network health and Array's pick performance across the fleet at customer sites.
ROS	Controller-board telemetry into ROS, for lighting control.
CI	State-machine diagrams generated in CI, so the architecture docs can't fall out of sync with the code.

Robots lose contact during in-space assembly, so my planner has no central coordinator



t = 2 s t = 6 s t = 12 s

fig. 2 Priority-Based Search moving 40 agents (0–39) through a cluttered grid.

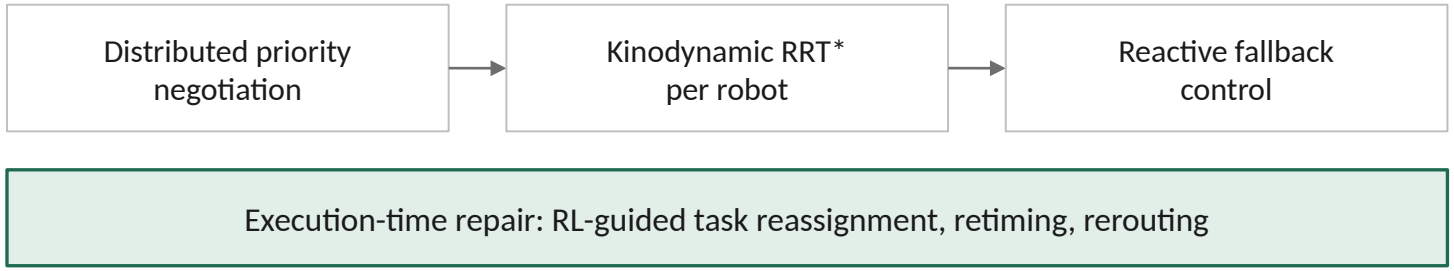


fig. 3 Planner layers. Repair triggers: comm loss, sensor drift, partially known maps.

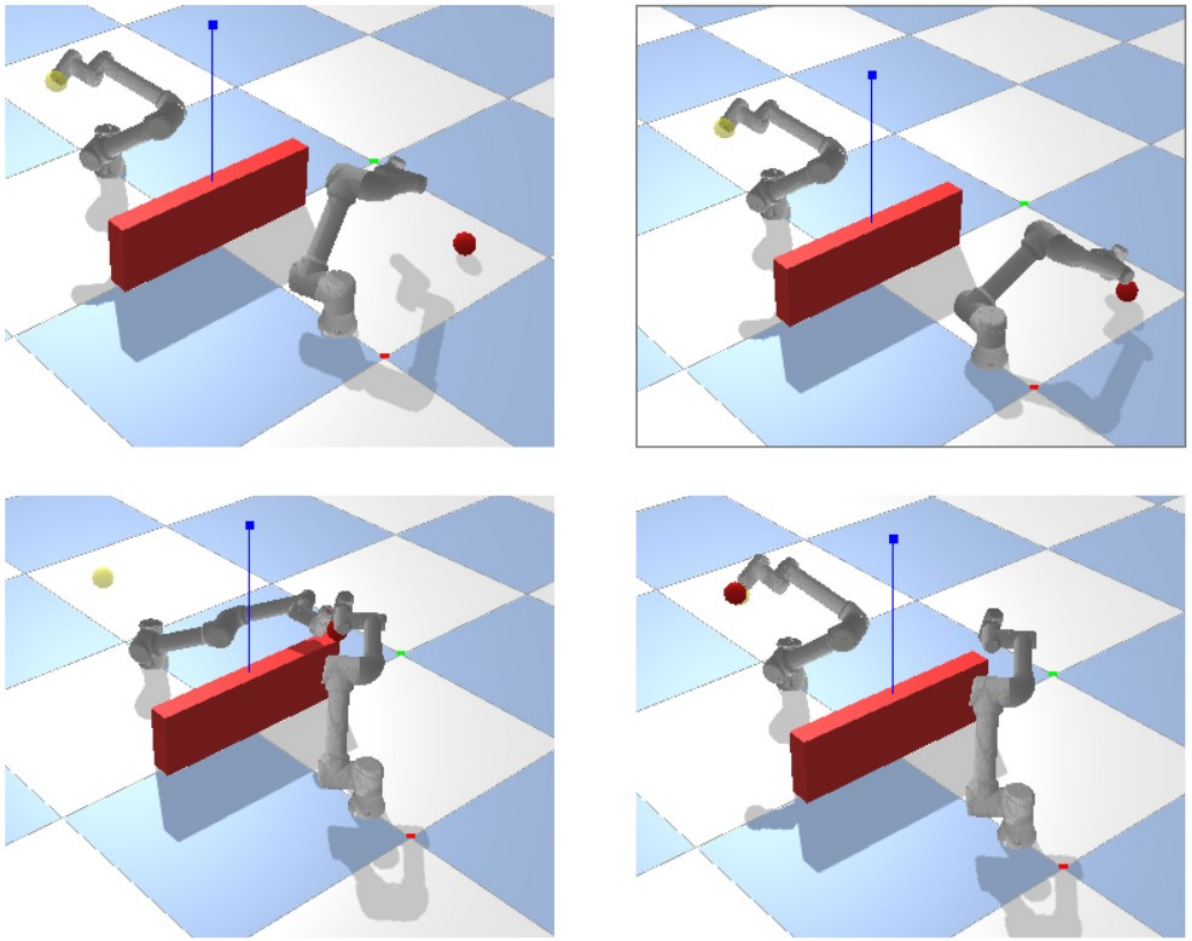
DESIGN CHOICES

- Robots negotiate priority with neighbors instead of waiting on a central planner that a dropped link would cut off.
- Paths come from kinodynamic RRT* because a free-flyer carries momentum and can't stop on command.
- When the plan can't be trusted, a reactive controller takes over.

STATUS

I'm designing the three planner layers and building the repair system (green) now.

Two UR5e arms pass a ball across a wall without touching it or each other



I implemented four sampling-based planners in C++ for arms with up to 7 DoF, then used them for a collision-avoidance system that coordinates two UR5e arms through a pick, a handoff, and a place in PyBullet.

PLANNERS

RRT · RRT-Connect · RRT* · PRM

fig. 4 (left) Read across, then down: start, pick, handoff over the wall, place. Red = ball, yellow = target.

Code: github.com/bmc209/16350-Final-Project

Reset vector to servo PWM: the Cortex-M4 kernel I'm writing register by register

I'm building a bare-metal kernel for an STM32 Cortex-M4. **I write every layer between the reset vector and user programs**: boot code, register-level drivers, interrupts, and syscalls. I use it to drive the robot car in fig. 6.

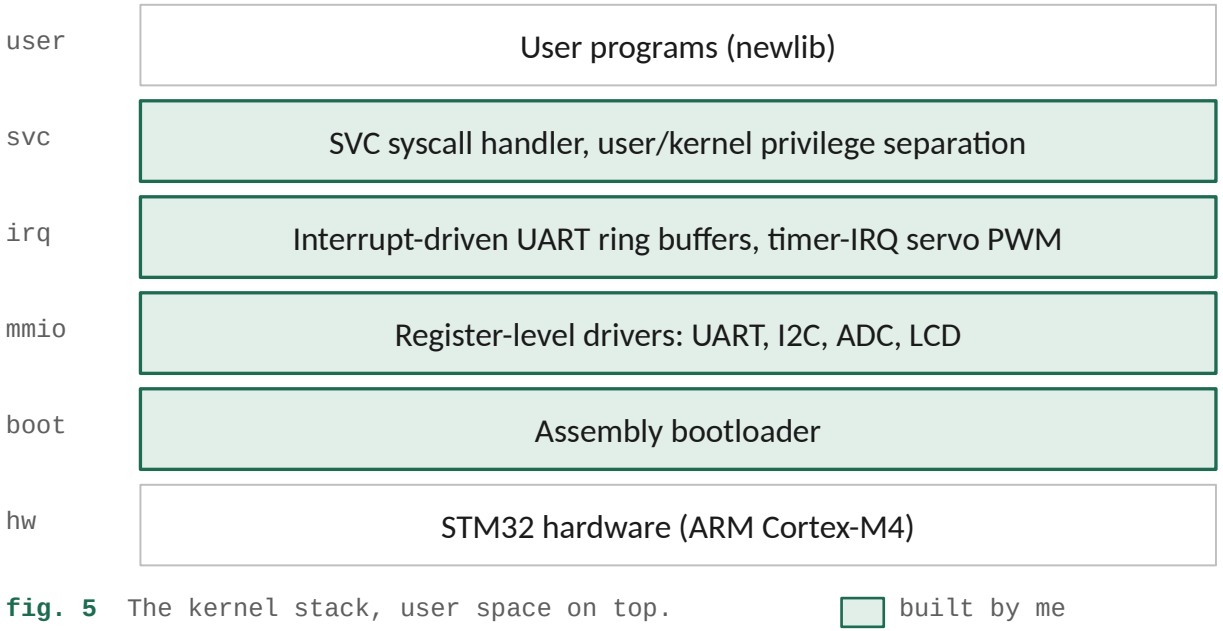


fig. 5 The kernel stack, user space on top.

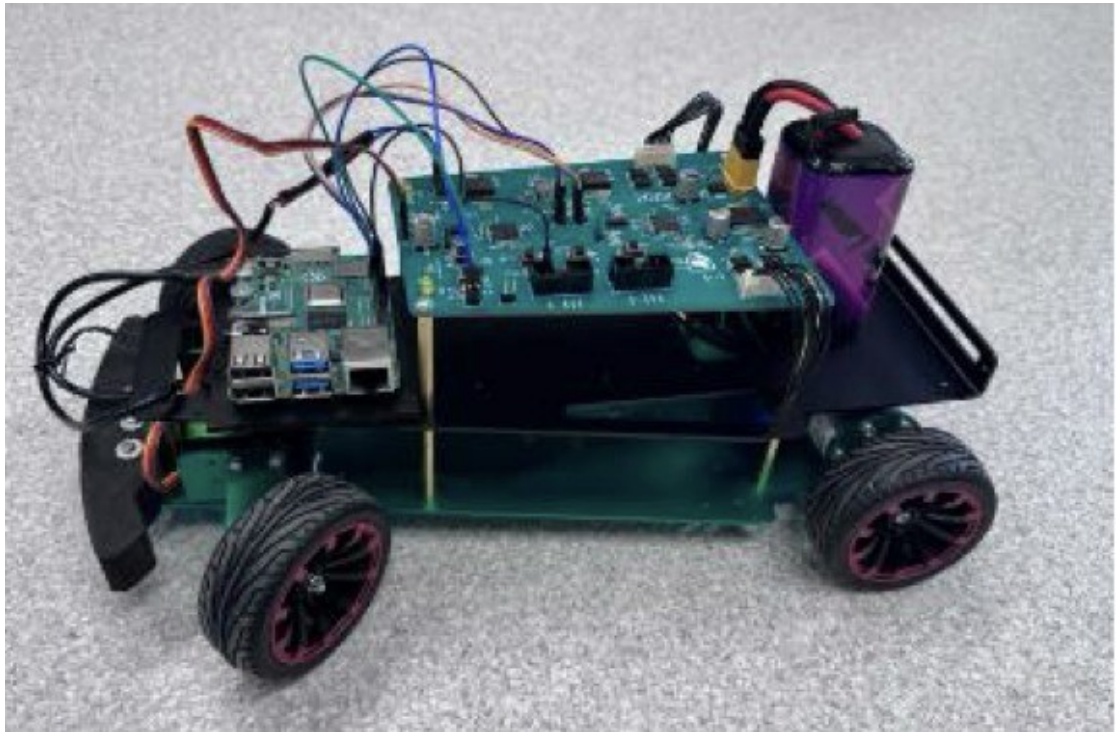


fig. 6 The 18-349 robot car.

Game logic as an FSM and a 12-bit datapath, deterministic to the nanosecond at 40–200 MHz

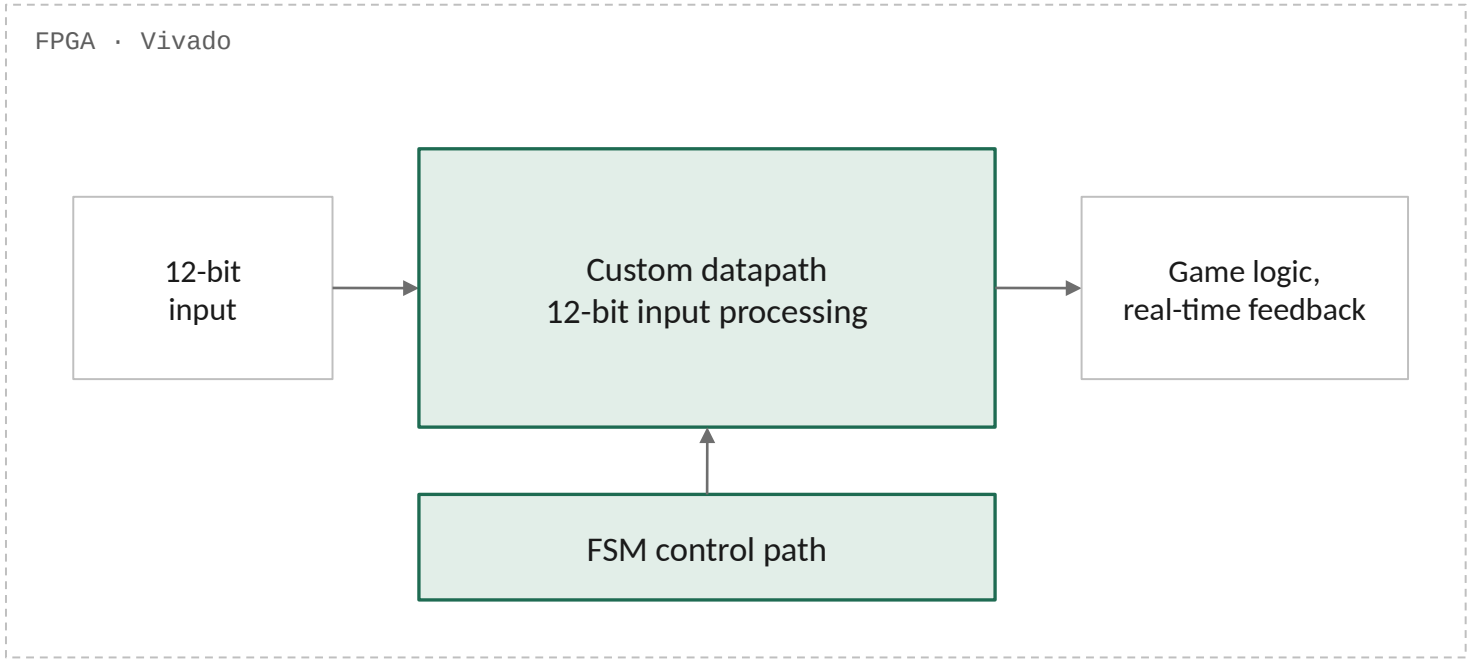


fig. 7 Block diagram, SystemVerilog. built by me

Designed in SystemVerilog and built in Vivado for high-speed game logic and real-time feedback on the FPGA.

Five applications built on a 3D print that senses its own bending

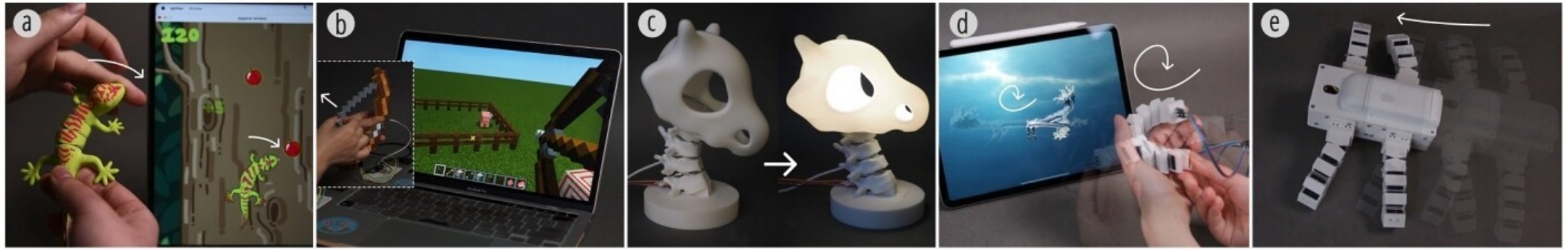


fig. 8 The five Xspine applications, a–e. Figure from the CHI '26 paper.

Xspine embeds motion sensing directly into flexible multi-material 3D prints, with no assembly. **I modeled their tactile and sensing feedback in software and built the pipeline that turns joint signals into training data** for AI models.

MY CONTRIBUTION

- Developed 5 application interfaces (fig. 8 a–e) that turn 8 joint signals streamed at 100 Hz into live game and toy input
- Virtually modeled force-tunable game controllers and fidget toys (Python, Unity, React)

PUBLISHED AS

Xspine: Integrating Motion Sensing Capability into Dynamic Structures Using Multi-material FDM 3D Printing.
ACM CHI 2026, full paper and interactive demo.
doi.org/10.1145/3772318.3791317

Integrating Motion Sensing into 3D-Printed Bending Structures.
ACM UIST 2025 Adjunct, poster.
doi.org/10.1145/3746058.3758350

I teach circuits to 130+ students a semester and lead builds for a team that placed 5th of 54

Jan 2026–now

Teaching Assistant (Lab Lead), 18-220 Electronic Devices and Analog Circuits

130+ students per semester. I lead 2 weekly recitations and 3-hour labs, hold office hours, and debug circuits with students at the bench.

Jan 2025–now

Senior Build Lead, ChemE Cube

I lead weekly build sessions for our Direct Air Capture system. 5th of 54 teams at the 2025 AIChE Annual Conference.

Jan 2025–now

SWE++ Curriculum Lead, Society of Women Engineers

An 8-week Python camp for 20+ middle schoolers. I write the lessons and starter code and manage the volunteers.

Sep 2026–now

Software & Perception, NASA Lunabotics (CMU team)

Joined this fall.